

Task 1: Understanding Cyber Security Basics & Attack Surface

1. Understanding Cybersecurity and the CIA Triad

Cybersecurity is about protecting computers, networks, apps, and data from attacks or unauthorized access.

It ensures that information and digital systems stay safe, correct, and available whenever we need them.

A very important idea in cybersecurity is the CIA Triad: Confidentiality, Integrity, and Availability. Confidentiality means keeping information private so only the right people can see it.

Example: Your bank account details are secret. Banking apps use passwords and two-factor authentication (2FA) to protect them.

Integrity means data stays correct and isn't changed by someone else without permission.

Example: On social media, no one can edit your messages or posts without your approval.

Systems also use checks like hashes to make sure messages are not tampered with.

Availability means the system and data are accessible when needed.

Example: Online banking should be available 24/7. If hackers perform a DoS attack, the service could be down, stopping you from using it.

2. Types of Attackers

Hackers or attackers have different skill levels and reasons for attacking.

Script Kiddies: Beginners who use ready-made tools to hack systems.

Example: Someone using software to hack Wi-Fi passwords without really understanding it.

Insiders: People who already have access, like employees, and misuse it.

Example: A bank employee leaks customer information.

Hacktivists: People who attack systems for political or social reasons.

Example: Hacking a government website to protest a law or policy.

Nation-State Actors: Professional hackers supported by governments.

Example: Hacking another country's defense systems or banks to gather intelligence.

3. Common Attack Surfaces

An attack surface is any place where a hacker can try to break into a system.

Common attack surfaces include:

Web Applications: Websites like banking portals. Hackers can use attacks like SQL injection or cross-site scripting (XSS).

Mobile Apps: Apps can leak personal information if they are not properly encrypted.
APIs: APIs allow apps to talk to each other. If they're insecure, hackers can steal data.
Networks: Unsecured Wi-Fi or poorly configured routers can be exploited.
Cloud Infrastructure: Poorly configured cloud servers can expose sensitive information online.

4. OWASP Top 10 Vulnerabilities

OWASP lists the top 10 web application security risks. These are common ways hackers exploit systems:

Broken Access Control: Users access things they shouldn't.
Cryptographic Failures: Sensitive data isn't encrypted properly.
Injection: Hackers insert malicious code into the system.
Insecure Design: Poorly designed systems are easy to attack.
Security Misconfiguration: Default passwords or open ports give hackers access.
Vulnerable Components: Using old or unsafe software libraries.
Identification & Authentication Failures: Weak login or session systems.
Software & Data Integrity Failures: Hackers tamper with updates or code.
Security Logging & Monitoring Failures: Attacks go unnoticed.
Server-Side Request Forgery (SSRF): Hackers trick the server into making requests.
These vulnerabilities can affect the confidentiality, integrity, or availability of data.

5. Mapping Daily Applications to Attack Surfaces

Here's how everyday apps can be attacked:

App	Attack Surface	Vulnerabilities
Gmail	Web, Mobile, API	Phishing, stealing login credentials, cross-site scripting
WhatsApp	Mobile App, Network, API	Malware, infected links, leaking private data
Banking Apps	Web, Mobile, API, Cloud	SQL injection, weak authentication, insecure storage

6. Data Flow: User → Application → Server → Database

How data moves when we use apps:

User: You type in your details (like a username or password).
Application: The app takes your data and sends it to the server.
Server: Processes the data and talks to the database to check or store information.
Database: Stores or retrieves the data.

Server → Application → User: The result is sent back to your device.

So the complete flow is:

User → Application → Server → Database → Server → Application → User

7. Possible Attack Points in This Flow

Hackers can attack at many points:

User Device: Malware or phishing attacks can steal your data.

Application Layer: Weak input forms or authentication can be exploited.

Server Layer: Misconfigured servers or code flaws can allow attacks.

Database Layer: SQL injections or unencrypted storage can expose data.

Network Layer: Hackers can intercept data in transit (man-in-the-middle attacks).

8. Summary

Cybersecurity protects data so it stays private (confidential), correct (integrity), and available when needed.

Attackers range from beginners to government-backed professionals. They target apps, networks, and cloud systems.

OWASP Top 10 vulnerabilities highlight common weaknesses that hackers exploit.

Daily apps like Gmail, WhatsApp, and banking apps have many attack surfaces, and attacks can happen at any stage of data flow.

Knowing how these attacks happen helps people and organizations protect their data better.

Interview Question

1)The CIA Triad is the foundation of cybersecurity. It stands for Confidentiality, Integrity, and Availability. Confidentiality means keeping information private so that only authorized people can access it. For example, your bank account password should only be known to you and not shared with anyone else. Integrity means keeping information accurate and unchanged. For instance, a hacker should never be able to change the amount in your bank account. Availability means that systems and data should always be accessible when needed. For example, you should be able to use online banking anytime you want, and it should not go down because of a cyberattack. In simple terms, confidentiality is about privacy, integrity is about accuracy, and availability is about access when needed.

2)An attack surface is the total number of possible points or areas where a hacker can try to enter or attack a system. You can think of it like a house — the more doors and windows it has, the more ways a thief can break in. Similarly, every login page, mobile app, API connection,

Wi-Fi network, or cloud storage service adds to a system's attack surface. The main goal of cybersecurity is to reduce the attack surface by securing every possible entry point so that hackers have fewer chances to break in.

3)Vulnerability, threat, and risk are three related but different terms in cybersecurity. A vulnerability is a weakness or flaw in a system. For example, using an outdated version of Windows that has known security bugs is a vulnerability. A threat is something or someone that can take advantage of that weakness, such as a hacker who finds that bug. Risk is the possible harm or damage that can happen if the threat successfully uses the vulnerability. For instance, if the hacker manages to steal your personal data, that is the risk becoming real. In simple words, vulnerability is the weakness, threat is the attacker, and risk is the impact or damage caused.

4)There are different types of cyber attackers. Script kiddies are beginners who use ready-made hacking tools without having much technical knowledge. Insiders are people who already have access to the system, such as employees, and misuse it for personal gain or revenge. Hacktivists are hackers who attack systems for political or social reasons, like protesting government policies. Cybercriminals are hackers who do it for money, such as stealing bank or credit card information. Nation-state actors are skilled hackers supported by governments who target other countries' important systems. In some rare cases, even competitors might try to hack or steal trade secrets to get an advantage in business.

5)OWASP, which stands for the Open Web Application Security Project, is a global organization that researches how hackers attack web applications. The OWASP Top 10 is a list of the ten most dangerous and common web application security risks. It is very important because it helps developers and security professionals understand what the biggest security problems are and how to fix or prevent them. The OWASP Top 10 is recognized all over the world as an industry standard in cybersecurity. It guides developers to build safer websites and applications by avoiding common issues such as weak passwords, injection attacks, and misconfigurations.